

Chase Hoffman

Software Developer | High-Performance C++ & Systems

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TECHNICAL SKILLS

Languages: C++ (14/17/20), Python, Lua, Rust (working knowledge)

Automotive & Embedded: Adaptive AUTOSAR, MISRA C++, Linux, RTOS (QNX), CAN, DOIP

Systems & Communication: TCP/IP, Ethernet, JSON-RPC, Protobuf, IPC, transport-agnostic architectures

Architecture & Concurrency: Multithreaded systems, event-driven design, lock-free data structures, low-latency systems

Tooling: CMake, Docker, GDB, Valgrind, Perf, Jenkins, GitHub Actions, GoogleTest

PROFESSIONAL EXPERIENCE

General Motors | Senior Embedded Software Engineer

April 2025 – Present

- Architected and owned a C++ edge telemetry engine with embedded Lua execution for dynamic runtime logic, enabling remote orchestration and rule deployment via Azure IoT Hub.
- Owned core C++ telemetry component integrating Boost lock-free SPSC queues for high-frequency ingestion and deterministic low-latency processing under concurrency.
- Collaborated on a Yocto-based Linux image for Raspberry Pi targets and integrated system-level components for embedded deployment.
- Architected a transport-agnostic C++ IPC framework using JSON-RPC to decouple vehicle compute components from underlying transport layers for consistent inter-component communication.

General Motors | Software Developer – Autonomous Driver Assist Platform (ADAS)

June 2023 – April 2025

- Optimized SoC startup latency by profiling dependency instantiation and refactoring system initialization sequence.
- Engineered persistent Key-Value storage in the Run-Time Environment (RTE) for configuration retention across ignition cycles.
- Owned and maintained system configuration daemon handling ingestion, validation, and lifecycle management of vehicle calibration and configuration parameters.

General Motors | Software Developer – Infotainment Platforms

June 2022 – June 2023

- Integrated SOME/IP and Protobuf stacks for communication between infotainment and safety-critical domains.
- Developed unit and integration tests for a distributed data middleware system (Zenoh), validating message flow and communication reliability across components.
- Contributed to Rust-based SOME/IP to custom protocol conversion components focused on memory safety and robustness.
- Supported CI/CD pipelines using Jenkins and GitHub Actions and maintained GoogleTest-based unit testing infrastructure.

Diversified Automation | Electrical Engineer

Aug 2018 – May 2020

- Designed industrial control systems and automated electrical schematic generation via Python, reducing manual engineering effort.

TECHNICAL PROJECTS

Terminal-Based Performance Profiler (C++) | In Progress

- CLI tool analyzing runtime behavior including lock contention, scheduling jitter, and memory churn in multithreaded systems.

Computer Vision Chess Clock (C++ / Raspberry Pi)

- Built real-time board state detection system using OpenCV to trigger automated chess timing events.

EDUCATION

M.S. in Computer Science | University of Colorado Boulder In Progress

B.A. in Electrical and Computer Engineering | University of Louisville 2021